

# Spatial Diversity in Radars—Models and Detection Performance

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**Abstract**—Inspired by recent advances in multiple-input multiple-output (MIMO) communications, this proposal introduces the statistical MIMO radar concept. To the authors' knowledge, this is the first time that the statistical MIMO is being proposed for radar. The fundamental difference between statistical MIMO and other radar array systems is that the latter seek to maximize the coherent processing gain, while statistical MIMO radar capitalizes on the diversity of target scattering to improve radar performance. Coherent processing is made possible by highly correlated signals at the receiver array, whereas in statistical MIMO radar, the signals received by the array elements are uncorrelated. Radar targets generally consist of many small elemental scatterers that are fused by the radar waveform and the processing at the receiver, to result in echoes with fluctuating amplitude and phase. It is well known that in conventional radar, slow fluctuations of the target radar cross section (RCS) result in target fades that degrade radar performance. By spacing the antenna elements at the transmitter and at the receiver such that the target angular spread is manifested, the MIMO radar can exploit the spatial diversity of target scatterers opening the way to a variety of new techniques that can improve radar performance. This paper focuses on the application of the target spatial diversity to improve detection performance. The optimal detector in the Neyman–Pearson sense is developed and analyzed for the statistical MIMO radar. It is shown that the optimal detector consists of noncoherent processing of the receiver sensors' outputs and that for cases of practical interest, detection performance is superior to that obtained through coherent processing. An optimal detector invariant to the signal and noise levels is also developed and analyzed. In this case as well, statistical MIMO radar provides great improvements over other types of array radars.

**Index Terms**—Array processing, multiple-input multiple-output (MIMO), radar.

## I. INTRODUCTION

**R**ADAR theory has been a vibrant scientific field for the last 50 years or so [1]–[3]. Radar theory deals with many different and diverse problems. However, the two most important problems are the detection and range estimation problems. The importance of these two problems is not limited to radars, and other engineering disciplines like sonar and communication deal with very similar problems [4]. Over the years, radar systems have developed considerably. These developments can be attributed to the increase in computation power and advances in hardware design. While early radar systems utilized a directional antenna, today's array radar systems can synthesize beams and simultaneously scan the whole space [5].

In array radars, the system is composed of many closely spaced omnidirectional antennas that either transmit or receive signals. Nonetheless, in some radar systems, only one antenna transmits energy [6], [7]. In active radar systems, the transmitting antennas transmit a known waveform. This waveform is reflected back from the target toward the receiving array. The radar system's tasks are to detect the existence of the target and to estimate its unknown parameters, e.g., range, speed, and direction.

In active array applied to radar systems, solutions for the detection and parameter estimation problems are well documented in the literature (see the survey in [8]). These solutions can be divided into two groups. The first group of solutions is based on high-resolution techniques, e.g., MUSIC or maximum-likelihood (ML) [8]. In the second group of solutions, an array of sensors is used to steer a beam toward a certain direction in space, in a manner similar to a conventional radar with a directional antenna [5]. The advantages of array radars are well known (see, e.g., [5], [8]–[11]). The lack of any mechanical elements in the system and the ability to steer multiple beams at once are two examples of such advantages.

Systems with more than one transmitting element have been proposed and built as well [9]. In these systems, an array of transmitting elements is used. In the conventional approach, an array of closely spaced transmitting elements is used to cohere a beam toward a certain direction in space, a process called *beam-forming* [5]. The main advantage of these systems is the lack of any mechanical elements, which reduces the system's complexity and improves its performance.

Recently, a new and interesting concept in array radar has been introduced by the synthetic impulse and aperture radar (SIAR) [12]–[14]. In SIAR systems, the elements of the transmitting array emit orthogonal waveforms. Through synthetic pulse formation, SIAR achieves the advantages of wide-band

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radar (improved range resolution) while individual antennas transmit narrow-band waveforms. Unlike conventional beamformers, SIAR features isotropic radiation (an advantage in terms of the probability of intercept of the radar waveform by a third party). Like beamformers, however, SIAR exploits the full correlation of signals transmitted and received at the array elements. Radars utilizing multiple transmitters and orthogonal waveforms have also been proposed in [15] and [16], where they have been referred to as multiple-input multiple-output (MIMO) radars.

The performance of radar systems is limited by target scintillations [17]. Targets are complex bodies composed of many scatterers. The range to, and the orientation of, the target determines the amount of energy reflected from these scatterers, and small changes in range or orientation can result in a large increase or decrease in the amount of energy reflected from the target [4], [17]. Both experimental measurements and theoretical results demonstrate that scintillations of 10 dB or more in the reflected energy can occur by changing the target aspect by as little as one milliradian [2]. These scintillations are responsible for signal fading, which can cause a large degradation in the system's performance.

Target scintillations are characterized by Swerling's target models. Most, if not all, radar processing ignores target scintillations and considers target scintillations as an unattended. Target radar cross section (RCS) fading can reduce the received signal energy to a level that does not allow reliable detection. One way to mitigate the effect of target fading is to maximize the received energy from the target. This, in turn, implies that one should maximize the system's coherent processing gain. Since processing gain can be realized with closely spaced antennas, most systems use an array of closely spaced antennas. A different approach addressing the target fading problem is to use diversity mechanisms. Frequency and polarity diversity may provide some relief, but in this paper, we focus on spatial diversity, an area that received very little attention for radar.

Motivated by recent breakthroughs in communications theory, in particular in MIMO systems and space-time coding [18]–[20], we question the conventional wisdom of maximizing the coherent processing gain. In this paper, we demonstrate that there are alternative optimization criteria that outperform beamforming and lead to a new radar architecture. In particular, we demonstrate that the utility of an array of sensors in which the spacing between the elements of the array is very large and orthogonal waveforms are transmitted from the elements of the transmitting array. The key point in the radar architecture is that sensors both at the transmitter and the receiver of the radar are separated such that they experience an *angular spread*.<sup>1</sup> As will be seen in the sequel, such a system forms a multistatic radar [21]. The advantage of this system is that the average received energy (averaged across all the independent radars) is approximately constant, i.e., it does not fade as in conventional systems. In essence, we will show that spatial diversity gain outweighs the coherent processing gain. We will demonstrate that an additional advantage of such systems is that the sensor outputs are combined noncoherently. While the idea of mul-

tistatic radars is not new, our approach is very different from the conventional approach, which assumes that the channel and clutter are deterministic unknowns. This results in detectors that try to estimate these unknown parameters as part of the detection algorithm used by the system. However, in this paper we develop a probabilistic model for the unknown parameters. We then take the Bayesian approach and derive the optimal detectors under different settings.

Communication systems that use the same principle for improving their performance are called MIMO systems. Since our proposed system is inspired by MIMO communications, and it exploits that statistical properties of the target RCS, we refer to it as *statistical MIMO radar*. The reader should note that the whole notion of statistical MIMO radar is new, and the main purpose of this paper is to introduce this concept and to demonstrate its advantages. For clarity and mathematical tractability, we use a simple model that ignores Doppler effects and clutter. More realistic models are left to subsequent work.

The rest of the paper is organized as follows. Section II develops a signal model that generalizes current signal models. This model is then used to classify different array radars and in particular to describe our proposed statistical MIMO radar. Section III analyzes the performance of the proposed radar, while Section IV examines the optimal invariance detector. Section V provides the summary and concluding remarks.

## II. SIGNAL MODEL

### A. General Theory

The point source assumption dominates the models used in radar theory [2]. For closely spaced sensors and large range between the target and the array, the point source assumption is an excellent approximation. However, as has been noted in the array signal processing community, a more accurate model is the distributed source model [22]–[24]. Unlike the common point source model, the distributed source model accounts for the spatial characteristics of the target. In statistical MIMO radar, the spacing between the array elements is large. Due to the target's complex shape and the distance between the array elements, every element observes a different aspect of the target. Therefore, the point source model is not adequate for describing the received signal in statistical MIMO radar, and a more detailed model must be developed.

One of the main contribution of this paper is to extend the classical Swerling-I model, to multiple-antenna systems. Consider a radar target, for example a ship. This target is composed from many small scatterers that are distributed over an area denoted by  $S$ . We can define a function  $\sigma(x, y)$  which defines the RCS of the scatterer located at  $x, y$ . Note that  $\sigma(x, y)$  is zero outside  $S$ . In reality,  $\sigma(x, y)$  has the form  $\sigma(x, y) = \sum_{i=1}^N \alpha_i \delta(x - x_i) \delta(y - y_i)$ , where  $\alpha_i$  is a realization of a random number. This model corresponds to the assumption that the target is composed from a finite but large number of scatterers distributed.

Unfortunately, the actual target physical model is not mathematically tractable. Swerling [4] and others (see, among many others, [22]) have taken a somewhat simplifying approach. They assumed that  $N = \infty$ ,  $S$  is either circular or rectangular and

<sup>1</sup>Angular spread is the target's RCS variability as a function of aspect ratio.

that the  $\alpha_i$ 's are realizations of some random process. In this paper, we take a similar approach. The reader should note that this simplifying assumptions do not compromise the accuracy of the model.

In what follows, we assume that the target is located at some point  $(x_0, y_0)$  in space, and it is stationary during the observation time, i.e., it conforms to the Swerling-I model [2]. The target has a rectangular shape whose dimensions are  $\Delta x \times \Delta y$ . In addition, we assume that our system is composed of  $M$  transmitters and  $N$  receivers. With reference to Fig. 1, we denote by  $(tx_k, ty_k), k = 1, \dots, M$  the locations (in the two-dimensional space) of the  $M$  transmitters and by  $(rx_l, ry_l), l = 1, \dots, N$  the locations of the  $N$  receivers. In the figure,  $t_1$  and  $t_2$  represent transmitters, while  $r_1$  and  $r_2$  represent the receivers.

We assume that the target is composed of an infinite number of random, isotropic and independent scatterers, uniformly distributed over  $[x_0 - (\Delta x/2), x_0 + (\Delta x/2)] \times [y_0 - (\Delta y/2), y_0 + (\Delta y/2)]$ . We denote by  $\Sigma(x, y)$  the complex gain of the scatterer located at  $(x + x_0, y + y_0)$ , for  $(x, y) \in [-(\Delta x/2), (\Delta x/2)] \times [-(\Delta y/2), (\Delta y/2)]$ . We model  $\Sigma(x, y)$  as a zero-mean, white, complex random variable, and  $E\{|\Sigma(x, y)|^2\} = (1/\Delta x \Delta y)$ . The last assumption is responsible for normalizing to one the average energy returned from the target.

In order to describe the received signal model, we need the following definitions: we denote by  $d(x, y, x', y') \triangleq \sqrt{(x - x')^2 + (y - y')^2}$  the distance between  $(x, y)$  and  $(x', y')$ . Let  $\tau(x, y, x', y') \triangleq (d(x, y, x', y')/c)$  be the time it takes a signal transmitted from  $(x, y)$  to reach  $(x', y')$ , where  $c$  is the speed of light. Assume that a narrow-band signal, denoted by  $\sqrt{(E/M)}s_k(t)$ , is transmitted from the  $k$ th transmitter, where  $\|s_k(t)\|^2 = 1$  and  $E$  is the total average received energy. The received signal at the  $l$ th receiver is the superposition of the signals reflected from all the scatterers. Denote by  $r_{lk}(t)$  the received signal at the  $l$ th receiver due to the signal transmitted from the  $k$ th transmitter, where  $r_{lk}(t)$  obeys

$$r_{lk}(t) = \int_{x_0 - \frac{\Delta x}{2}}^{x_0 + \frac{\Delta x}{2}} \int_{y_0 - \frac{\Delta y}{2}}^{y_0 + \frac{\Delta y}{2}} \sqrt{\frac{E}{M}} s_k(t - \tau(tx_k, ty_k, \gamma, \beta) - \tau(rx_l, ry_l, \gamma, \beta)) \Sigma(\gamma - x_0, \beta - y_0) d\gamma d\beta.$$

A change of variables  $\beta + x_0 \rightarrow \beta$  and  $\gamma + y_0 \rightarrow \gamma$  results in the following expression for  $r_{lk}(t)$ :

$$r_{lk}(t) = \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} \sqrt{\frac{E}{M}} s_k(t - \tau(tx_k, ty_k, x_0, y_0) - (\tau(tx_k, tx_k, x_0 + \gamma, y_0 + \beta) - \tau(tx_k, ty_k, x_0, y_0)) - \tau(rx_l, ry_l, x_0, y_0) - (\tau(rx_l, ry_l, x_0 + \gamma, y_0 + \beta) - \tau(rx_l, ry_l, x_0, y_0))) \Sigma(\gamma, \beta) d\gamma d\beta.$$

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$$r_{lk}(t) = \sqrt{\frac{E}{M}} s_k(t - \tau(tx_k, ty_k, x_0, y_0) - \tau(rx_l, ry_l, x_0, y_0)) \times \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-j2\pi f_c (\tau(tx_k, ty_k, x_0 + \gamma, y_0 + \beta) - \tau(tx_k, tx_k, x_0, y_0) + \tau(rx_l, ry_l, x_0 + \gamma, y_0 + \beta) - \tau(rx_l, ry_l, x_0, y_0))} \Sigma(\gamma, \beta) d\gamma d\beta. \quad (1)$$

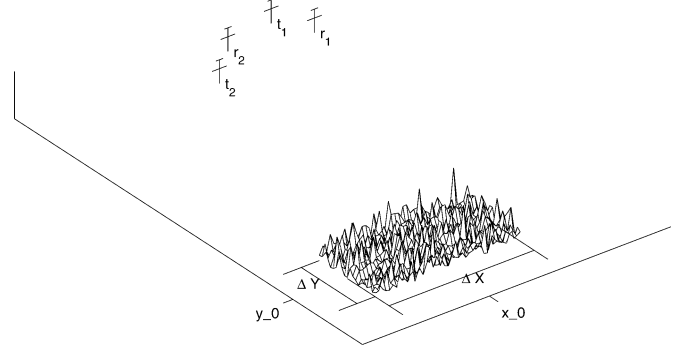


Fig. 1. Spatial configuration.

Invoking the narrow-band assumption, we have (1), shown at the bottom of the page, where  $f_c$  is the carrier frequency. This model can be further simplified by using the approximations in (2), shown at the bottom of the next page, where the first approximation is due to the fact that  $\gamma^2 + \beta^2 \ll (tx_k - x_0)^2 + (ty_k - y_0)^2$ , and the second is due to  $\sqrt{x + \epsilon} \approx \sqrt{x} + (\epsilon/2\sqrt{x})$  for  $\epsilon \ll x$ .

By combining (1) and (2), and using the narrow-band signal assumption, the received signal can be modeled as (3), shown at the bottom of the next page, where  $\alpha_{lk}$  is the gain between the  $k$ th transmitter and the  $l$ th (see (4), shown at the bottom of the next page).

Since  $\Sigma(x, y)$  is a random field,  $\alpha_{lk}$  is a random variable. The exact distribution of  $\alpha_{lk}$  depends on the exact distribution of  $\Sigma(x, y)$ . However, due to the central limit theorem,  $\alpha_{lk}$  is approximately a complex normal random variable. The mean and variance of  $\alpha_{lk}$  are given in (5) and (6), shown at the bottom of the next page, where we used the fact that  $E\{\Sigma(\gamma, \beta) \Sigma^*(\eta, \zeta)\} = (1/\Delta x \Delta y) \delta(\gamma - \eta) \delta(\beta - \zeta)$ . Consequently, the distribution of  $\alpha_{lk}$  can be approximated as  $\alpha_{lk} \sim \mathcal{CN}(0, 1)$ .

In order to write the received signal in compact-matrix form, we need additional notation. Let

$$\psi_k = 2\pi f_c (\tau(tx_k, ty_k, x_0, y_0) - \tau(tx_1, ty_1, x_0, y_0)) \quad (7)$$

$$\varphi_l = 2\pi f_c (\tau(rx_l, ry_l, x_0, y_0) - \tau(rx_1, ry_1, x_0, y_0)). \quad (8)$$

Since the signal is narrow-band,

$$s(t - \tau(tx_k, ty_k, x_0, y_0) - \tau(rx_l, ry_l, x_0, y_0)) = e^{-j\psi_k - j\varphi_l} s(t - \tau(tx_1, ty_1, x_0, y_0) - \tau(rx_1, ry_1, x_0, y_0));$$

and the received signal model, (3), can be further simplified as follows:

$$r_{lk}(t) = \alpha_{lk} e^{-j\psi_k - j\varphi_l} \sqrt{\frac{E}{M}} s_k(t - \tau(tx_1, tx_1, x_0, y_0) - \tau(rx_1, ry_1, x_0, y_0)). \quad (9)$$

The received signal at the  $l$ th receiver is the superposition of all the signals originating from the various transmitters plus

the additive noise. Denote by  $r_l(t)$  the received signal and by  $n_l(t)$  the additive noise at the  $l$ th receiver and then  $r_l(t)$  by the following model:

$$r_l(t) = \sqrt{\frac{E}{M}} \sum_{k=1}^M \alpha_{lk} e^{-j\psi_k - j\varphi_l} s_k(t - \tau) + n_l(t) \quad (10)$$

where  $\tau \triangleq \tau(tx_1, tx_1, x_0, y_0) - \tau(rx_1, ry_1, x_0, y_0)$ .

$$\begin{aligned} & \tau(tx_k, ty_k, x_0 + \gamma, y_0 + \beta) - \tau(tx_k, ty_k, x_0, y_0) \\ &= \frac{\sqrt{(tx_k - (x_0 + \gamma))^2 + (ty_k - (y_0 + \beta))^2} - \sqrt{(tx_k - x_0)^2 + (ty_k - y_0)^2}}{c} \\ &= \frac{\sqrt{(tx_k - x_0)^2 + \gamma^2 + 2\gamma(tx_k - x_0) + (ty_k - y_0)^2 + \beta^2 + 2\beta(ty_k - y_0)} - \sqrt{(tx_k - x_0)^2 + (ty_k - y_0)^2}}{c} \\ &\approx \frac{\sqrt{(tx_k - x_0)^2 + 2\gamma(tx_k - x_0) + (ty_k - y_0)^2 + 2\beta(ty_k - y_0)} - \sqrt{(tx_k - x_0)^2 + (ty_k - y_0)^2}}{c} \\ &\approx \frac{\sqrt{(tx_k - x_0)^2 + (ty_k - y_0)^2} + \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{\sqrt{(tx_k - x_0)^2 + (ty_k - y_0)^2}} - \sqrt{(tx_k - x_0)^2 + (ty_k - y_0)^2}}{c} \\ &= \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{c\sqrt{(tx_k - x_0)^2 + (ty_k - y_0)^2}} = \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{c \cdot d(tx_k, ty_k, x_0, y_0)}. \end{aligned} \quad (2)$$

$$\begin{aligned} r_{lk}(t) &= \sqrt{\frac{E}{M}} s_k(t - \tau(tx_k, ty_k, x_0, y_0) - \tau(rx_l, ry_l, x_0, y_0)) \\ &\quad \cdot \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-2j\pi f_c \left( \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{c \cdot d(tx_k, ty_k, x_0, y_0)} + \frac{\beta(rx_l - x_0) + \gamma(ry_l - y_0)}{c \cdot d(rx_l, ry_l, x_0, y_0)} \right)} \Sigma(\gamma, \beta) d\gamma d\beta \\ &= \alpha_{lk} \sqrt{\frac{E}{M}} s_k(t - \tau(tx_k, ty_k, x_0, y_0) - \tau(rx_l, ry_l, x_0, y_0)). \end{aligned} \quad (3)$$

$$\alpha_{lk} \triangleq \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-2j\pi f_c \left( \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{c \cdot d(tx_k, ty_k, x_0, y_0)} + \frac{\beta(rx_l - x_0) + \gamma(ry_l - y_0)}{c \cdot d(rx_l, ry_l, x_0, y_0)} \right)} \Sigma(\gamma, \beta) d\gamma d\beta. \quad (4)$$

$$\begin{aligned} & E\{\alpha_{lk}\} \\ &= \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-2j\pi f_c \left( \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{c \cdot d(tx_k, ty_k, x_0, y_0)} + \frac{\beta(rx_l - x_0) + \gamma(ry_l - y_0)}{c \cdot d(rx_l, ry_l, x_0, y_0)} \right)} E\{\Sigma(\gamma, \beta)\} d\gamma d\beta = 0, \\ & E\{|\alpha_{lk}|^2\} \\ &= \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-j2\pi f_c \left( \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{c \cdot d(tx_k, ty_k, x_0, y_0)} + \frac{\beta(rx_l - x_0) + \gamma(ry_l - y_0)}{c \cdot d(rx_l, ry_l, x_0, y_0)} - \frac{\eta(tx_k - x_0) + \zeta(ty_k - y_0)}{c \cdot d(tx_k, ty_k, x_0, y_0)} - \frac{\eta(rx_l - x_0) + \zeta(ry_l - y_0)}{c \cdot d(rx_l, ry_l, x_0, y_0)} \right)} \end{aligned} \quad (5)$$

$$\begin{aligned} & E\{\Sigma(\gamma, \beta)\Sigma^*(\eta, \zeta)\} d\gamma d\beta d\eta d\zeta \\ &= \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} \frac{1}{\Delta x \Delta y} d\beta d\gamma = 1. \end{aligned} \quad (6)$$

Denote by  $\mathbf{r}(t) = [r_1, \dots, r_N(t)]^T$  the collection of the received signals at the various receiving elements, and by  $\mathbf{s}(t) = [s_1(t), \dots, s_M(t)]^T$  the collection of the transmitted signals from the various transmitting elements. The received signal vector can be described by the following simple model:

$$\mathbf{r}(t) = \sqrt{\frac{E}{M}} \text{diag}(\mathbf{a}(x_0, y_0)) \mathbf{H} \text{diag}(\mathbf{b}(x_0, y_0)) \mathbf{s}(t - \tau) + \mathbf{n}(t) \quad (11)$$

where  $\mathbf{H}$  is an  $N \times M$  matrix, referred to as the channel matrix such that  $[\mathbf{H}]_{ji} = \alpha_{ji}$ ;  $\text{diag}(\mathbf{v})$  is a diagonal matrix with  $\mathbf{v}$  on its diagonal;  $\mathbf{a}(x_0, y_0) = [1, e^{i\varphi_2}, \dots, e^{i\varphi_N}]^T$  is an  $N \times 1$  vector, which is a function of the target location, usually referred to as the receiver steering vector;  $\mathbf{b}(x_0, y_0) = [1, e^{i\psi_2}, \dots, e^{i\psi_M}]^T$  is an  $M \times 1$  vector, which is a function of the target location, usually referred to as the transmitter steering vector; and  $\mathbf{n}(t) = [n_1(t), \dots, n_N(t)]^T$  is an  $N \times 1$  vector representing the additive noise. We assume that  $\mathbf{n}(t)$  is a white, zero-mean, complex normal random process with correlation matrix  $\sigma_n^2 \mathbf{I}_N$ , where  $\mathbf{I}_N$  is the  $N \times N$  matrix.

In order to complete the received signal model, (11), we need to characterize the distribution of the channel matrix  $\mathbf{H}$ . As will be seen in the sequel, the exact distribution of the channel matrix depends on the distances between the array elements and the target. Different array-target configurations give rise to different channel matrix distributions. The MIMO<sup>2</sup> concept is related to one configuration, whereas the conventional model is related to another. Before we discuss these different configurations, we analyze the distribution of the channel matrix further.

We have already demonstrated that every element of the channel matrix  $\mathbf{H}$  is a zero-mean, unit variance complex normal random variable. Denote by  $\boldsymbol{\alpha} = [\alpha_{11}, \dots, \alpha_{1M}, \alpha_{21}, \dots, \alpha_{NM}]^T$  the vector that contains all the elements of the matrix  $\mathbf{H}$ . It is clear from our previous discussion that  $\boldsymbol{\alpha}$  is complex normal random vector. Since the

mean of each element of  $\boldsymbol{\alpha} \sim \mathcal{CN}(\mathbf{0}_{MN}, \mathbf{R}_{\boldsymbol{\alpha}})$ , where  $\mathbf{0}_{MN}$  is the  $MN \times 1$  all-zero vector, and  $\mathbf{R}_{\boldsymbol{\alpha}} = E\{\boldsymbol{\alpha}\boldsymbol{\alpha}^H\}$  (the superscript  $\cdot^H$  denotes conjugate transpose) is the correlation matrix of the vector  $\boldsymbol{\alpha}$ . The matrix  $\mathbf{R}_{\boldsymbol{\alpha}}$ , depends on the exact arrays-target configuration. However, there exists several cases in which we can approximate  $\mathbf{R}_{\boldsymbol{\alpha}}$  using very simple expressions.

Consider  $\alpha_{jk}$  and  $\alpha_{jl}$ . The correlation between  $\alpha_{jk}$  and  $\alpha_{jl}$  is given by (12), shown at the bottom of the page, where we used the fact that  $E\{\Sigma(x, y)\Sigma(u, v)\} = (1/\Delta x \Delta y)\delta(x-u)\delta(y-v)$ .

In order to get further insight into  $E\{\alpha_{jk}\alpha_{jl}^*\}$ , we develop conditions under which  $|E\{\alpha_{jk}\alpha_{jl}^*\}|$  is either approximately zero or approximately one. Define  $d \triangleq (d(tx_k, ty_k, x_0, y_0) + d(tx_l, ty_l, x_0, y_0))/2$  and  $d' \triangleq (d(tx_k, ty_k, x_0, y_0) - d(tx_l, ty_l, x_0, y_0))/2$ , and consider

$$\begin{aligned} & \frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} e^{-j\frac{2\pi}{\lambda_c} \left( \frac{\beta(tx_k - x_0)}{\tau(tx_k, ty_k, x_0, y_0)} - \frac{\beta(tx_l - x_0)}{\tau(tx_l, ty_l, x_0, y_0)} \right)} d\beta \\ &= \frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} e^{-j\frac{2\pi}{\lambda_c} \frac{(d-d')(tx_k - x_0) - (d+d')(tx_l - x_0)}{(d-d')(d+d')}} \beta d\beta \quad (13) \end{aligned}$$

where  $\lambda_c$  is the carrier wavelength. It is easily seen that a sufficient condition for the above expression to be much less than one is that the integrand completes at least one cycle or equivalently  $(2\pi/\lambda_c) ((d-d')(tx_k - x_0) - (d+d')(tx_l - x_0))/(d-d')(d+d') \frac{\Delta x}{2} > \pi$ . Using some algebraic manipulations, this condition can be reduced to

$$(tx_k - tx_l)(d - d') + 2d'x_0 > \lambda_c(d + d')(d - d')/\Delta x. \quad (14)$$

Without loss of generality, we can choose  $d'$  to have the same sign as  $x_0$ , hence  $2d'x_0 > 0$ , and we can strengthen the inequality by requiring that

$$d' = (tx_k - tx_l) > \lambda_c(d + d')/\Delta x. \quad (15)$$

$$\begin{aligned} E\{\alpha_{jk}\alpha_{jl}^*\} &= E \left\{ \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-2j\pi f_c \left( \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{c \cdot d(tx_k, ty_k, x_0, y_0)} + \frac{\beta(rx_j - x_0) + \gamma(ry_j - y_0)}{d \cdot d(rx_j, ry_j, x_0, y_0)} \right)} \Sigma(\gamma, \beta) \right. \\ &\quad \cdot e^{j\frac{2\pi}{\lambda_c} \left( \frac{\eta(tx_l - x_0) + \xi(ty_l - y_0)}{d(tx_l, ty_l, x_0, y_0)} + \frac{\eta(rx_j - x_0) + \xi(ry_j - y_0)}{d(rx_j, ry_j, x_0, y_0)} \right)} \Sigma(\eta, \xi) d\gamma d\beta d\eta d\xi \Big\} \\ &= \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-j\frac{2\pi}{\lambda_c} \left( \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{d(tx_k, ty_k, x_0, y_0)} + \frac{\beta(rx_j - x_0) + \gamma(ry_j - y_0)}{d(rx_j, ry_j, x_0, y_0)} - \frac{\eta(tx_l - x_0) + \xi(ty_l - y_0)}{d(tx_l, ty_l, x_0, y_0)} - \frac{\eta(rx_j - x_0) + \xi(ry_j - y_0)}{d(rx_j, ry_j, x_0, y_0)} \right)} \\ &\quad \cdot E\{\Sigma(\gamma, \beta)\Sigma(\eta, \xi)\} d\gamma d\beta d\eta d\xi \\ &= \frac{1}{\Delta x \Delta y} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-j\frac{2\pi}{\lambda_c} \left( \frac{\beta(tx_k - x_0) + \gamma(ty_k - y_0)}{d(tx_k, ty_k, x_0, y_0)} + \frac{\beta(rx_j - x_0) + \gamma(ry_j - y_0)}{d(rx_j, ry_j, x_0, y_0)} - \frac{\beta(tx_l - x_0) + \gamma(ty_l - y_0)}{d(tx_l, ty_l, x_0, y_0)} - \frac{\beta(rx_j - x_0) + \gamma(ry_j - y_0)}{d(rx_j, ry_j, x_0, y_0)} \right)} d\gamma d\beta \\ &= \frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} e^{-j\frac{2\pi}{\lambda_c} \left( \frac{\beta(tx_k - x_0)}{d(tx_k, ty_k, x_0, y_0)} - \frac{\beta(tx_l - x_0)}{d(tx_l, ty_l, x_0, y_0)} \right)} d\beta \frac{1}{\Delta y} \int_{-\frac{\Delta y}{2}}^{\frac{\Delta y}{2}} e^{-j\frac{2\pi}{\lambda_c} \left( \frac{\gamma(ty_k - y_0)}{d(tx_k, ty_k, x_0, y_0)} - \frac{\gamma(ty_l - y_0)}{d(tx_l, ty_l, x_0, y_0)} \right)} d\gamma. \quad (12) \end{aligned}$$

<sup>2</sup>For readability, in the sequel, we shorten “statistical MIMO” to plain “MIMO.”

Condition (15) has a simple physical interpretation. The target is illuminated by the transmitter and it reflects the energy back. The target can be regarded as an antenna with aperture  $\Delta x$  and beamwidth  $\lambda_c/\Delta x$ . If two transmitters are not within the same receive beamwidth of the target, then they see different “aspects” of the target with uncorrelated RCSs. Our condition examines whether the two transmitters are separated by more than the angular spread of the target or not.

This simple condition, (15), allows us to determine whether the elements of the channel matrix  $\mathbf{H}$  considered correlated or not. Consider, for example, the  $jk$ th and  $il$ th elements of the channel matrix  $\mathbf{H}$ . If at least one of the following four conditions:

$$\begin{aligned} (1) \quad & rx_j - rx_i > d(rx_j, ry_j, x_0, y_0)\lambda_c/\Delta x \\ (2) \quad & tx_k - tx_l > d(tx_k, ty_k, x_0, y_0)\lambda_c/\Delta x \\ (3) \quad & ry_j - ry_i > d(rx_j, ry_j, x_0, y_0)\lambda_c/\Delta y \\ (4) \quad & ty_k - ty_l > d(tx_k, ty_k, x_0, y_0)\lambda_c/\Delta y \end{aligned} \quad (16)$$

holds, the  $jk$ th and  $il$ th elements of the channel matrix are *uncorrelated*. Note that (16) is a sufficient condition.

In contrast to (16), we would like to develop a similar condition under which two elements of the channel matrix  $\mathbf{H}$  are deemed correlated. Based on the discussion so far, if the integrands in (13) do not complete more than, say, 10% of a cycle, the correlation between  $\alpha_{jk}$  and  $\alpha_{il}$  is close to one. It can be easily seen from (16) that if the following four conditions hold jointly:

$$\begin{aligned} (1) \quad & rx_j - rx_i \ll d(rx_j, ry_j, x_0, y_0)\lambda_c/\Delta x \\ (2) \quad & tx_k - tx_l \ll d(tx_k, ty_k, x_0, y_0)\lambda_c/\Delta x \\ (3) \quad & ry_j - ry_i \ll d(rx_j, ry_j, x_0, y_0)\lambda_c/\Delta y \\ (4) \quad & ty_k - ty_l \ll d(tx_k, ty_k, x_0, y_0)\lambda_c/\Delta y \end{aligned} \quad (17)$$

then the  $jk$ th and  $il$ th elements of the channel matrix are approximately *fully correlated*.

## B. Model Classification

Our received signal model (11) can be used to describe many different systems. These systems differ in two aspects: the correlation between the elements of the channel matrix  $\mathbf{H}$  and the design of the transmitted signals. In this paper, we consider four canonical systems, which we believe represent the four extreme cases. These four systems are the conventional phased-array radar, the MIMO radar, and the multiple-input single-output (MISO) and the single-input multiple-output (SIMO) radars. Each of these four canonical systems represents a large number of similar systems. Consequently, it is sufficient to examine these systems in order to compare which one is preferable and where.

1) *Phased-Array Radar*: Phased-array radars are probably the most common radar systems that use an array of sensors. Phased-array radars utilizes an array of closely spaced sensors. That is, every pair of elements obeys condition (17). In addition, in phased-array radar, the transmitted signal is given by  $\mathbf{s}(t) = \tilde{\mathbf{b}}\mathbf{s}(t)$ , where  $\tilde{\mathbf{b}}$  is usually referred to as the transmitter steering vector.

When every pair of elements obeys condition (17), all the elements of the channel matrix are fully correlated. Therefore, the channel matrix is given by  $\alpha\mathbf{1}_{NM}$ , where  $\alpha \sim \mathcal{CN}(0, 1)$ , and  $\mathbf{1}_{NM}$  is the  $N \times M$  all-ones matrix. Particularizing the received signal model (11) to this case results in the following received signal:

$$\mathbf{r}(t) = \sqrt{\frac{E}{M}}\alpha\mathbf{a}(x_0, y_0)\mathbf{b}(x_0, y_0)^H\tilde{\mathbf{b}}\mathbf{s}(t - \tau) + \mathbf{n}(t). \quad (18)$$

In phased-array radar systems, both  $\mathbf{a}(x_0, y_0)$  and  $\mathbf{b}(x_0, y_0)$  are functions of the angle between the array and the target. It is common to denote these steering vectors by  $\mathbf{a}(\theta)$  and  $\mathbf{b}(\theta')$ , respectively, where  $\theta$  is the angle between the receiving (transmitting) array and the target. If, in addition, the receiver uses a beamformer to steer toward directions  $\tilde{\theta}$ , and the transmitter steering vector  $\tilde{\mathbf{b}} = \mathbf{b}(\theta')$ , then the output of the beamformer is

$$\begin{aligned} r(t) &= \mathbf{a}(\tilde{\theta})^H\mathbf{r}(t) \\ &= \sqrt{\frac{E}{M}}\alpha\mathbf{a}(\tilde{\theta})^H\mathbf{a}(\theta)\mathbf{b}(\theta')^H\mathbf{b}(\tilde{\theta})\mathbf{s}(t - \tau) + n(t). \end{aligned} \quad (19)$$

This model represents a bistatic radar where  $\mathbf{b}(\theta')^H\mathbf{b}(\tilde{\theta})$  plays the role of the transmit-antenna pattern, and  $\mathbf{a}(\theta)^H\mathbf{a}(\tilde{\theta})$  the receive-antenna pattern. Since  $n(t) = \mathbf{a}(\tilde{\theta})^H\mathbf{n}(t)$  is a linear transformation of  $\mathbf{n}(t)$ ,  $n(t)$  is a zero-mean, complex normal random process with correlation function  $\|\mathbf{a}(\tilde{\theta})\|^2\sigma_n^2\delta(\tau) = N\sigma_n^2\delta(\tau)$ .

In phased-array radar, a processing gain of  $NM$  can be realized by taking  $\theta = \tilde{\theta}$  and  $\theta' = \tilde{\theta}'$ ,  $\mathbf{a}(\tilde{\theta})^H\mathbf{a}(\theta) = N$ , and  $\mathbf{b}(\tilde{\theta})^H\mathbf{b}(\theta) = M$ . This processing gain does not come free of cost. Phased-array radars are sensitive to the distribution of the fading coefficient  $\alpha$ . If it happens that  $\alpha$  is small, there is no way to overcome this fade, and the target will not be detected, or the estimation error will be large. MIMO radars are capable of overcoming this exact problem by exploiting spatial diversity.

The radars proposed in [15] has been referred to by the authors as MIMO radars. In these radars, each transmitter transmits one of  $M$  orthogonal waveforms. It was shown in [15] that with proper processing, phased-array radars that use this type of signaling can realize a coherent processing gain of  $N$ . However, since the transmitted signals are orthogonal, this radar has a low probability of intercept (LPI). According to our classification, these radars belong to the phased-array category.

2) *MIMO Radar*: While in phased-array radar, the interelement spacing between every pair of elements obeys condition (17), in MIMO radar the interelement spacings between each pair of elements obeys condition (16). Therefore, all the elements of the channel matrix are uncorrelated, that is  $\alpha \sim \mathcal{CN}(0, \mathbf{I}_{MN})$ . Particularizing the received signal model (11) to this case results in the following model:

$$\mathbf{r}(t) = \sqrt{\frac{E}{M}}\mathbf{H}\mathbf{s}(t - \tau) + \mathbf{n}(t). \quad (20)$$

It should be noted that since  $\text{diag}(\mathbf{a}(x_0, y_0))$  and  $\text{diag}(\mathbf{b}(x_0, y_0))$  in (11) are diagonal matrices with elements on the unit circle, the distribution of  $\text{diag}(\mathbf{a}(x_0, y_0))\mathbf{H}\text{diag}(\mathbf{b}(x_0, y_0))$  equals the distribution of  $\mathbf{H}$ .

With MIMO radar, each transmitter–receiver pair sees a different aspect of the target. In particular, by proper selection of the transmitted signals from the various transmitters, one can generate the equivalent of  $MN$  radar systems.

Assume that each of the  $M$  transmitters transmits one of  $M$  signals, and denote these signals by  $s_i(t)$ ,  $i = 1, \dots, M$ . By match-filtering each of the elements of  $\mathbf{r}(t)$  with each of the transmitted signals, we can reconstruct  $r_{ji}(t) = \sqrt{(E/M)}\alpha_{ji}s_i(t - \tau) + n_{ji}(t)$ , where  $n_{ji}(t)$  is a zero-mean, complex normal random process with correlation function  $R_n(\tau) = \sigma_n^2\delta(\tau)$ .

While in conventional radar the processing gain is  $MN$  and the radar sees only single aspect of the target, in MIMO radar, there is no coherent processing gain, but we synthesize  $MN$  independent radars. As a result, MIMO radar overcomes deep fades.

3) *MISO and SIMO Radar*: In conventional radar, all the interelement spacings are small, whereas in MIMO radar, elements are widely spaced. It is sometime desirable to mix these two approaches. MISO and SIMO radars are a compromise between phased-array radars and MIMO radars. In MISO radar, the interelement spacings of the transmitting elements obey (16), while the interelement spacings of the receiving elements obey (17). In contrast, in SIMO radar, the interelement spacings of the transmitting elements obey (17), while the interelement spacings of the receiving elements obey (16). In what follows, we concentrate on the MISO radars, and we omit the discussion on SIMO radars due to space limitations.

Since every pair of transmitting elements obeys (16),  $\alpha_{ji}$  and  $\alpha_{kl}$  are uncorrelated for  $i \neq l$ . However, since every pair of receiving elements obeys (17),  $\alpha_{ji}$  and  $\alpha_{ki}$  are fully correlated. Therefore, the channel matrix  $\mathbf{H}$  equals  $\mathbf{1}_N\boldsymbol{\alpha}^H$ , where  $\boldsymbol{\alpha} \sim \mathcal{CN}(0, \mathbf{I}_M)$ , and  $\mathbf{1}_N$  is the all-ones vector of length  $N$ . Particularizing the received signal model (11) to this case results in the following model:

$$\mathbf{r}(t) = \sqrt{\frac{E}{M}}\mathbf{a}(\theta)\boldsymbol{\alpha}^H\mathbf{s}(t - \tau) + \mathbf{n}(t). \quad (21)$$

MISO radar can realize a coherent processing gain of  $\|\mathbf{a}(\theta)\|^2 = N$  and create  $M$  independent radars. To see this, suppose that, as in MIMO radar, each of the  $M$  transmitters transmits one of  $M$  orthogonal signals  $s_i(t)$  ( $i = 1, \dots, M$ ). The receiver first steers toward  $\theta$  and then uses a beamformer to combine the received signal vectors. By match-filtering the beamformed signal, we can construct the following  $M$  received signals:  $r_i(t) = N\sqrt{(E/M)}\alpha_i s_i(t - \tau) + n_i(t)$ , where  $n_i(t)$  is a zero-mean, complex normal random process with correlation function  $R_n(\tau) = N\sigma_n^2\delta(\tau)$ . A complete analysis of MISO radars and their application for direction finding can be found in [25].

### III. BASIC PERFORMANCE COMPARISONS

The radar detection problem can be formulated as follows [17]:

$$\begin{aligned} \mathcal{H}_0: & \text{Target does not exist at delay } \tau \\ \mathcal{H}_1: & \text{Target exists at delay } \tau. \end{aligned} \quad (22)$$

Many variants of this basic detection problem have been investigated and analyzed in the past. These variants differ by the assumed signal model, the unknown parameters, etc. In this section, we investigate the best achievable performance with phased-array, MIMO, and MISO radars. We then compare the various systems and determine the optimal one.

Since we are interested in the inherent limitations of the various systems, in this section, we assume that  $\theta, \theta'$ , and the noise level  $\sigma_n^2$  are all known in advance. The optimal, in the Neyman–Pearson sense, detector is the likelihood ratio test (LRT), which is given by [26]

$$T = \log \frac{f(\mathbf{r}(t) | \mathcal{H}_1)}{f(\mathbf{r}(t) | \mathcal{H}_0)} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{>}} \delta \quad (23)$$

where  $f(\mathbf{r}(t) | \mathcal{H}_0)$  and  $f(\mathbf{r}(t) | \mathcal{H}_1)$  are the probability density functions (pdf) of the observation vector given the null and alternate hypotheses, respectively, and  $\delta$  is a threshold, set by the desired probability of false alarm.

#### A. MIMO Radar

The following Lemma determines the structure of the MIMO radar's LRT detector.

*Lemma 1*: Denote by  $\mathbf{x}$  the  $NM \times 1$  vector such that  $[\mathbf{x}]_{iN+j} \triangleq \int r_i(t)s_j(t - \tau)dt$ , that is,  $\mathbf{x}$  is the output of a bank of matched filters. The optimal detector is given by

$$T = \|\mathbf{x}\|^2 \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{>}} \delta \quad (24)$$

where  $\delta$  is set to ensure the required probability of false alarm.

*Proof of Lemma 1*: See Appendix A.  $\square$

Notice the noncoherent nature of the MIMO radars detector. By proper processing, the detector create  $MN$  virtual conventional radar systems and then it sums noncoherently the results of these systems. This is in contrast to the optimal detector of the phased-array system, to be described in the next subsection.

It is easy to verify that  $\mathbf{x}$  is distributed as follows:

$$\mathbf{x} = \begin{cases} \mathbf{n} & \mathcal{H}_0 \\ \sqrt{\frac{E}{M}}\boldsymbol{\alpha} + \mathbf{n} & \mathcal{H}_1 \end{cases} \quad (25)$$

where  $\mathbf{n} \sim \mathcal{CN}(0, \sigma_n^2\mathbf{I}_{NM})$  and  $\boldsymbol{\alpha} \sim \mathcal{CN}(0, \mathbf{I}_{NM})$ . Therefore,  $\mathbf{x}$  is a zero-mean complex random variable with correlation matrix  $\sigma_n^2\mathbf{I}_{NM}$  under the null hypothesis, and  $((E/M) + \sigma_n^2)\mathbf{I}_{NM}$  under the alternate hypothesis. This leads to the following distribution of the test statistic:

$$\|\mathbf{x}\|^2 \sim \begin{cases} \frac{\sigma_n^2}{2}\chi_{(2MN)}^2 & \mathcal{H}_0 \\ \left(\frac{E}{2M} + \frac{\sigma_n^2}{2}\right)\chi_{(2MN)}^2 & \mathcal{H}_1 \end{cases} \quad (26)$$

where  $\chi_{(d)}^2$  denotes a chi-square random variable with  $d$  degrees of freedom.

It is well known that the probability of false alarm, the probability of detection, and the threshold are all connected by a series of one-to-one relations. These relations can be written with the aid of (26). The probability of false alarm can be expressed as

$$\begin{aligned} \text{Pr}_{\text{FA}} &= \Pr(T > \delta | \mathcal{H}_0) = \Pr\left(\frac{\sigma_n^2}{2}\chi_{2MN}^2 > \delta\right) \\ &= \Pr\left(\chi_{2MN}^2 > \frac{2\delta}{\sigma_n^2}\right). \end{aligned} \quad (27)$$

It follows that  $\delta$  is set using the following formula:

$$\delta = \frac{\sigma_n^2}{2} F_{\chi_{2MN}^2}^{-1}(1 - \text{Pr}_{\text{FA}}) \quad (28)$$

where  $F_{\chi_{2MN}^2}^{-1}$  denotes the inverse cumulative distribution function of a chi-square random variable with  $2MN$  degrees of freedom. The probability of detection is given by

$$\begin{aligned} \text{Pr}_D &= \Pr(T > \delta | \mathcal{H}_1) = \Pr\left(\left(\frac{E}{2M} + \frac{\sigma_n^2}{2}\right) \chi_{2MN}^2 > \delta\right) \\ &= 1 - F_{\chi_{2M}^2}\left(\frac{2\delta}{\frac{E}{M} + \sigma_n^2}\right) \\ &= 1 - F_{\chi_{2MN}^2}\left(\frac{\sigma_n^2}{\frac{E}{M} + \sigma_n^2} F_{\chi_{2MN}^2}^{-1}(1 - \text{Pr}_{\text{FA}})\right). \end{aligned} \quad (29)$$

It is interesting to note that both the test statistic (24) and the threshold (28) are independent of the transmitted energy  $E$ . Therefore, the optimal detector, even in the case of unknown signal energy, is given by (24) and (28). This establishes the fact that (24) is a uniformly most powerful (UMP) detector for which only the noise level needs to be known [27]. In addition, closed-form formulas for (24), (28), and (29) exist and can be found in [28].

### B. Phased-Array Radar

Consider a phased-array radar system. The following lemma describes the structure of the phased-array LRT detector.

*Lemma 2:* Let  $x = \int \mathbf{r}^H(t) \mathbf{a}(\theta) s(t - \tau) dt$  be the output of the spatial-temporal matched filter. The optimal detector is given by

$$T = |x|^2 \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{>}} \delta \quad (30)$$

where  $\delta$  is set to ensure that the required probability of false alarm is achieved.

*Proof of Lemma 2:* See Appendix B.

The optimal detector for the phased-array system turns out to be the beamformer steering to direction  $\theta$  followed by a simple linear filter (also known as the Echardt filter) [29]. While the optimal detector is independent of the transmitter steering vector, the performance of the optimal detector depends on this vector. It is easily shown using the Cauchy–Schwartz inequality that the optimal performance is achieved when  $\hat{\mathbf{b}} = \mathbf{b}(\theta')$ . When this is the case, it is easy to see that  $x$  is described by the following model:

$$x = \begin{cases} n & \mathcal{H}_0 \\ \sqrt{\frac{E}{M}} \|\mathbf{a}(\theta)\|^2 \|\mathbf{b}(\theta')\|^2 \alpha + n & \mathcal{H}_1 \end{cases} \quad (31)$$

where  $\alpha \sim \mathcal{CN}(0, 1)$  and  $n \sim \mathcal{CN}(0, \sigma_n^2 \|\mathbf{a}(\theta)\|^2)$ . This gives rise to the following distribution of the test statistic:

$$T \sim \begin{cases} \frac{\sigma_n^2 N}{2} \chi_{(2)}^2 & \mathcal{H}_0 \\ \left(\frac{\sigma_n^2 N}{2} + \frac{EN^2 M^2}{2M}\right) \chi_{(2)}^2 & \mathcal{H}_1 \end{cases} \quad (32)$$

where we used the fact that  $\|\mathbf{a}(\theta)\|^2 = N$  and  $\|\mathbf{b}(\theta')\|^2 = M$ .

The probability of false alarm, the probability of detection, and the threshold can be easily derived using the same derivation

leading, respectively, to (27), (28), and (29). The probability of false alarm, the probability of detection, and the threshold are, respectively, given by

$$\text{Pr}_{\text{FA}} = \Pr\left(\chi_{(2)}^2 > \frac{2\delta}{\sigma_n^2 N}\right) \quad (33)$$

$$\delta = \frac{\sigma_n^2 N}{2} F_{\chi_{(2)}^2}^{-1}(1 - \text{Pr}_{\text{FA}}) \quad (34)$$

$$\text{Pr}_D = 1 - F_{\chi_{(2)}^2}\left(\frac{\sigma_n^2}{(\sigma_n^2 + ENM)} F_{\chi_{(2)}^2}^{-1}(1 - \text{Pr}_{\text{FA}})\right). \quad (35)$$

### C. MISO Radar

The last system we consider is the MISO radar. It is very easy to verify that MISO radar is a hybrid of both the MIMO and the phased-array systems. Therefore, it should not be a surprise that the optimal detector is a combination of the corresponding optimal detectors. The following lemma describes the MISO radar's optimal detector.

*Lemma 3:* Denote by  $\mathbf{x}$  an  $M \times 1$  vector such that  $[\mathbf{x}]_i \triangleq \int \mathbf{r}^H(t) \mathbf{a}(\theta) s_i(t - \tau) dt$ , that is,  $\mathbf{x}$  is the output of a bank of spatial-temporal matched filters. The optimal detector is then given by

$$T = \|\mathbf{x}\|^2 \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{>}} \delta \quad (36)$$

where  $\delta$  is set to ensure the required probability of false alarm.

The Proof of Lemma 3 is a simple modification of the proofs of Lemmas 1 and 2, and hence it is omitted. The receiving array is composed of closely spaced sensors. Consequently, the receiver can cohere a beam toward the target and coherently combine the signals received at the receiving array. However, the spatially combined signal contains several orthogonal signals sent from different antennas. The optimal detector despreads these orthogonal signals. Since these signals arrive with unknown gains and phases, the detector combines these signals noncoherently, the same way the MIMO radar does.

The probability of false alarm and the probability of detection can be computed from (27), (29), (33), and (35), and they are given respectively by

$$\text{Pr}_{\text{FA}} = \Pr\left(\chi_{(2M)}^2 > \frac{2\delta}{\sigma_n^2 \|\mathbf{a}(\theta)\|^2}\right) \quad (37)$$

$$\delta = \frac{\sigma_n^2 \|\mathbf{a}(\theta)\|^2}{2} F_{\chi_{(2M)}^2}^{-1}(1 - \text{Pr}_{\text{FA}}) \quad (38)$$

$$\text{Pr}_D = 1 - F_{\chi_{(2M)}^2}\left(\frac{\sigma_n^2}{(\sigma_n^2 + \frac{EN}{M})} F_{\chi_{(2M)}^2}^{-1}(1 - \text{Pr}_{\text{FA}})\right). \quad (39)$$

### D. Discussion

The most useful performance measure of any detector is its probability of detection. Therefore, we should compare the probability of detection of the various systems in order to determine the best. We will do this comparison based on numerical examples in the next section. Another approach, which is analytically tractable, is to compare the various systems based on a single scalar performance measure. Such a performance measure should summarize the detector performance in a single number instead of a function. One such performance measure

is the detector's signal-to-noise ratio (SNR) (also referred to as divergence in [30]).

Let  $T$  be some test statistic (detector). The detector's SNR, denoted by  $\beta$ , is defined as follows:

$$\beta = \frac{|E(T|\mathcal{H}_0) - E(T|\mathcal{H}_1)|^2}{\frac{1}{2}[\text{Var}(T|\mathcal{H}_0) + \text{Var}(T|\mathcal{H}_1)]}. \quad (40)$$

If  $T|\mathcal{H}_i, i = 0, 1$  is normally distributed,  $\sqrt{\beta}$  normalized distance between the means of the distribution of the detector's test statistics under the null and alternate hypotheses. This in turn represents our ability to distinguish between the two hypotheses. Although, in our problem,  $T|\mathcal{H}_i$  is not normally distributed, we find this measure very useful in capturing the detector's performance in one simple number.

Consider a MIMO radar system. According to (26),  $E(T|\mathcal{H}_0) = MN\sigma_n^2$ , and  $E(T|\mathcal{H}_1) = MN\sigma_n^2 + NE$ , and hence  $|E(T|\mathcal{H}_0) - E(T|\mathcal{H}_1)|^2 = N^2E^2$ . Also, according to (26),  $\text{Var}(T|\mathcal{H}_0) = MN\sigma_n^4$ , and  $\text{Var}(T|\mathcal{H}_1) = MN(E^2/M^2 + 2E\sigma_n^2 + \sigma_n^4)$ , and hence  $[\text{Var}(T|\mathcal{H}_0) + \text{Var}(T|\mathcal{H}_1)] = MN(E^2/M^2 + 2E\sigma_n^2/M + 2\sigma_n^4)$ . Combining these results, we obtain the following detector SNR:

$$\begin{aligned} \beta_{\text{MIMO}} &= \frac{E^2N^2}{MN(E^2/2M^2 + \sigma_n^4 + E\sigma_n^2/M)} \\ &= M(E^2/2M^2 + \sigma_n^4 + E\sigma_n^2/M). \end{aligned} \quad (41)$$

Define the SNR, denoted by  $\rho$ , as the ratio between the total transmitted energy and the noise level per receive element, that is  $\rho \triangleq E/\sigma_n^2$ . The detector SNR can be further simplified and is given by the following expression:

$$\beta_{\text{MIMO}} = \frac{\rho^2N}{M(1 + \rho^2/2M^2 + \rho/M)}. \quad (42)$$

Now consider a phased-array radar system. According to (32),  $E(T|\mathcal{H}_0) = N\sigma_n^2$ , and  $E(T|\mathcal{H}_1) = N\sigma_n^2 + N^2ME$ , and hence  $|E(T|\mathcal{H}_0) - E(T|\mathcal{H}_1)|^2 = N^4M^2E^2$ . Also, according to (32),  $\text{Var}(T|\mathcal{H}_0) = N^2\sigma_n^4$ , and  $\text{Var}(T|\mathcal{H}_1) = N^2\sigma_n^4 + E^2N^4M^2 + 2\sigma_n^2EN^3M$ , and hence  $[\text{Var}(T|\mathcal{H}_0) + \text{Var}(T|\mathcal{H}_1)] = 2N^2\sigma_n^4 + E^2N^4M^2 + 2\sigma_n^2EN^3M$ . The SNR of the optimal detector for the phased-array radar system is given by the following expression:

$$\begin{aligned} \beta_{\text{phased array}} &= \frac{N^4M^2E^2}{\frac{1}{2}[2N^2\sigma_n^4 + E^2N^4M^2 + 2\sigma_n^2EN^3M]} \\ &= \frac{N^2M^2E^2}{\frac{1}{2}[2\sigma_n^4 + E^2N^2M^2 + 2\sigma_n^2EMN]} \\ &= \frac{N^2M^2\rho^2}{1 + M^2N^2\rho^2/2 + N\rho M}. \end{aligned} \quad (43)$$

Consider now a MISO system. The SNR of the optimal detector for a MISO is given by the following expression (a complete derivation is omitted due to space limitations):

$$\begin{aligned} \beta_{\text{MISO}} &= \frac{E^2N^4}{\frac{1}{2}[2\sigma_n^4N^2M + 2N^3E\sigma_n^2 + \frac{E^2N^4}{M}]} \\ &= \frac{\rho^2N^2}{M + \rho^2N^2/M + N\rho} \\ &= \frac{\rho^2N^2}{M(1 + \rho^2N^2/2M^2 + N\rho/M)}. \end{aligned} \quad (44)$$

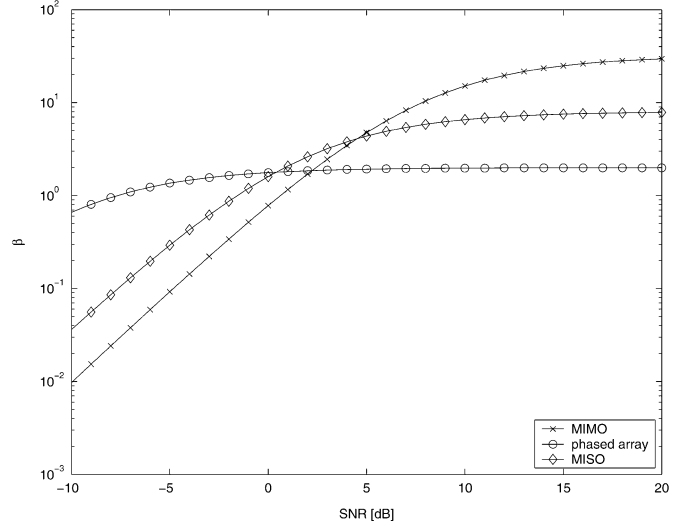


Fig. 2. Optimal detector's SNR in various systems.  $M = N = 4$ .

Fig. 2 depicts the optimal detectors' SNR for the phased-array, MIMO, and MISO systems. We assume that  $M = N = 4$  and that the noise level is constant across the array. It is evident from the figures that, while for high SNR the MIMO system is optimal, for low SNR the phased-array system is optimal.

Assume that  $\rho \gg 1$ , which is high SNR. The optimal detectors' SNR can then be approximated as follows:

$$\beta_{\text{MIMO}} \approx 2NM, \quad \beta_{\text{phased array}} \approx 2, \quad \beta_{\text{MISO}} \approx 2M. \quad (45)$$

It is easily seen that in terms of its ability to distinguish between the two hypotheses, the MIMO radar is the best system, whereas the phased-array system is the worst system. The received signal in a phased-array system enjoys a coherent processing gain. However, the random gain due to target fluctuations,  $\alpha$ , is Rayleigh distributed, and hence there is a nonnegligible probability that  $\alpha$  is either small or large. When  $\alpha$  is small, although the transmitted power is high, the received power is low. Therefore, the received SNR is low with non-negligible probability. The MIMO radar system, on the other hand, is equivalent to  $NM$  independent radar systems, each of which is subject to Rayleigh distributed random gain. Since the number of independent radars is large, some of them are subject to deep fades, while others experience strong gains. However, the average received power does not vary considerably. Therefore, the MIMO radar is not subject to the deep fades that affect the phased-array radar.

Assume that  $\rho \ll 1$ , which is low SNR. The optimal detectors' SNR can be approximated as follows:

$$\begin{aligned} \beta_{\text{MIMO}} &\approx \rho^2N/M, \quad \beta_{\text{phased array}} \approx \rho^2N^2M^2, \\ \beta_{\text{MISO}} &\approx 2M. \end{aligned} \quad (46)$$

For low SNR, the phased-array radar system's disadvantage becomes its advantage. Since with nonnegligible probability the random gain,  $\alpha$ , is large, in those instances, the instantaneous SNR is high compared with the average SNR. Consequently, the target is detected by the phased-array radar. On the other hand, the MIMO radar system does not enjoy this benefit. The received SNR can not deviate considerably from the average

received SNR (recall that we have  $MN$  independent radars). Consequently, the probability of detection of the MIMO radar system is lower than that of the phased-array system.

#### IV. THE OPTIMAL INVARIANT DETECTOR

In the previous section, we examined the performance of the optimal detector when all the parameters are known. This assumption is an unrealistic one, and usually both the noise level  $\sigma_n^2$  and the transmitted power  $E$  are unknown. In real-life systems, the noise level  $\sigma_n^2$  depends on the clutter, and the transmitted power  $E$  includes implicitly the unknown target average RCS. Consequently, it is unrealistic to assume that either  $E$  or  $\sigma_n^2$  are known in advance [4]. It should be noted that in many realistic scenarios, the noise level is different across the array and it might even be correlated. This case will be treated in subsequent paper.

In radar theory, there exist detectors that do not require explicit knowledge of either  $E$  or  $\sigma_n^2$ . These detectors are usually constant-false-alarm-rate detectors (CFAR), and some of them are optimal in a very important sense [17]. In what follows, we derive the optimal detector for MIMO and MISO systems when both the noise level  $\sigma_n^2$  and the transmitted energy  $E$  are unknown. We demonstrate that a very intuitive extension of the mean-level detector (MLD) is the optimal detector for MIMO radar systems in the invariant sense [31], [32].

When unknown parameters exist, many hypothesis testing problems do not possess a UMP detector. However, if we restrict our attention to a class of detectors having some special properties, e.g., CFAR, sometimes an optimal detector can be found. Usually these special properties are so natural that it is hard to imagine using a detector that does not possess these properties. The mathematical theory governing these special detectors is the theory of invariant decision rules. The theory of invariant decision rules is beyond the scope of this paper. (We refer the reader to [27], [32], and [33] for an introduction to invariant hypothesis testing theory.) In order to enhance an intuitive understanding of the subject, we describe the mathematical theory of invariant decision rules in the context of our detection problem. We first formulate the exact detection problem when both the noise level and the average received power are unknown and then develop the optimal detector in the invariance sense.

##### A. Problem Formulation

In all the systems discussed so far, the output of the matched filter sampled at  $t = \tau$  was a sufficient statistic. However, if both the noise level  $\sigma_n^2$  and the average received power  $E$  are unknown, the information contained at the output of the matched filter sampled at  $t = \tau$  is not sufficient for constructing a detector with bounded probability of false alarm. Take, for example, (25). Assume the null hypothesis  $\mathcal{H}_0$  and that  $\sigma_n^2 = 2$ . The output of the matched-filter sampled at  $t = \tau$  is a zero-mean, complex normal random vector with correlation matrix  $2\mathbf{I}$ . Assume, on the other hand, the alternate hypothesis  $\mathcal{H}_1$  and that  $\sigma_n^2 = 1$  and  $E = M$ . Again, the output of the matched filter sampled at  $t = \tau$  is a zero-mean, complex normal random vector with correlation matrix  $2\mathbf{I}$ . This simple example demonstrates that once both  $E$  and  $\sigma_n^2$  are unknown, we can find one

combination of the parameters that belongs to the null hypothesis, and another that belongs to the alternate hypothesis, which yield the same output of the matched filter sampled at  $t = \tau$ , i.e., the output is unidentifiable [34].

In order to overcome this problem, the common practice is to obtain additional samples of the noise process. This can be done, for example, by sampling the output of the matched filter in the vicinity of  $t = \tau$ . For a discussion on how to obtain these noise samples, we refer the reader to [17]. Here, we assume that  $L$  samples of the noise process were obtained by the receiver, and denote by  $\mathbf{y} = [y_1, \dots, y_L]^T$  the  $L \times 1$  vector that contains these samples, that is  $\mathbf{y} \sim \mathcal{CN}(0, \sigma_n^2 \mathbf{I}_L)$ .

Denote by  $\mathbf{x}$  the sufficient statistic when all the parameters are known, and by  $l(\mathbf{x}) = |\mathbf{x}|$  the length of  $\mathbf{x}$ . In addition, denote by  $\tilde{\mathbf{z}} \triangleq [\mathbf{y}^T, \mathbf{x}^T]^T$  the concatenation of  $\mathbf{y}$ . It is easy to verify that independent of the exact system, e.g., phased array, MIMO, MISO, used.

$$\tilde{\mathbf{z}} \sim \mathcal{CN}(0, \text{diag}([\sigma_n^2 \mathbf{1}_{L \times 1}, (\sigma_n^2 + \mu) \mathbf{1}_{l(\mathbf{x}) \times 1}])) \quad (47)$$

where  $\mu$  depends on the system considered and whether or not a target exists. It can be easily shown that  $\mathbf{z} = [||\mathbf{y}||^2, ||\mathbf{x}||^2]$  is a sufficient statistic for  $\tilde{\mathbf{z}}$ . Therefore, we will restrict our attention to  $\mathbf{z}$  instead of  $\tilde{\mathbf{z}}$ . The sufficient statistic vector  $\mathbf{z}$  is distributed as follows:

$$\mathbf{z} \sim \left[ \frac{\sigma_n^2}{2} \chi_{(2L)}^2, \frac{\sigma_n^2 + \mu}{2} \chi_{(2l(\mathbf{x}))}^2 \right]. \quad (48)$$

The detection problem can be formulated

$$\begin{aligned} \mathcal{H}_0: \mu &= 0, & \sigma_n^2 &\in \mathcal{R}_+ \\ \mathcal{H}_1: \mu &> 0, & \sigma_n^2 &\in \mathcal{R}_+ \end{aligned} \quad (49)$$

where  $\mathcal{R}_+$  is the set of positive, real numbers.

This detection problem does not possess a UMP detector. However, it is only natural to restrict our attention to detectors that are invariant to scaling, that is, detectors, the performance of which depends only on the SNR and not on the actual noise level and average received signal energy. Moreover, if the structure of this detector is independent of the exact SNR, then the resulting detector is what is known as an UMP invariant detector.

##### B. Invariant Decision Rules and the Optimal Detector

In this section, we provide a short introduction to invariant decision rules. We describe only the essential mathematical concepts without any specific proofs. After each step, we demonstrate the mathematical concept with our detection problem (49).

Let  $\mathbf{Z}$  be a random vector whose pdf, denoted by  $f_{\mathbf{Z}}(\mathbf{z}|\boldsymbol{\theta})$ ,  $\boldsymbol{\theta} \in \Theta$ , is known up to some unknown parameter vector, denoted by  $\boldsymbol{\theta}$ , belonging to a parameter space denoted by  $\Theta$ . We denote by  $\mathbf{z}$  the observation vector, and by  $\mathcal{Z}$  the observation space. Assume that  $\Theta = \Theta_0 \cup \Theta_1$  and  $\Theta_0 \cap \Theta_1 = \emptyset$ . We associate  $\Theta_0$  with the null hypothesis and  $\Theta_1$  with the alternate hypothesis, that is, if  $\mathcal{H}_0$  is true,  $\boldsymbol{\theta} \in \Theta_0$ , and if  $\mathcal{H}_1$  is true,  $\boldsymbol{\theta} \in \Theta_1$ . In our detection problem,  $\boldsymbol{\theta} = [\sigma_n^2, \mu]$  and  $\Theta = \mathcal{R}_+^2$ . In addition,  $\Theta_0 = (\mathcal{R}_+, 0)$ , and  $\Theta_1 = (\mathcal{R}_+, \mathcal{R}_+ - \{0\})$ .

Let  $\mathcal{G}$  denote a group of transformations from  $\mathbf{z}$  to  $\mathbf{z}$ . In addition, let the group operation be composition, that is, if  $g_1, g_2 \in \mathcal{G}$ , then  $g \triangleq g_2 g_1$  is the transformation that takes  $\mathbf{z}$  to  $g(\mathbf{z}) = g_2(g_1(\mathbf{z}))$ .

**Definition 1:** The family of distributions  $f_{\mathbf{Z}}(\mathbf{z} | \boldsymbol{\theta})$  is said to be invariant under the group  $\mathcal{G}$ , if for every  $g \in \mathcal{G}$  and  $\boldsymbol{\theta} \in \Theta$  there exists a unique  $\boldsymbol{\theta}' \in \Theta$  such that  $g(\mathbf{z})$  is distributed according to  $f(\mathbf{z} | \boldsymbol{\theta}')$ . Denote this  $\boldsymbol{\theta}'$  by  $\bar{g}(\boldsymbol{\theta})$ . The interpretation is that  $g$  has left the pdf invariant in form, but the parameter  $\boldsymbol{\theta}$  has changed to  $\bar{g}(\boldsymbol{\theta})$ .

Our main objective is to determine the optimal detector invariant to scaling, that is, the detector outcome when presented with both  $\mathbf{z}$  and  $a\mathbf{z}$  should be the same. Using this in our problem leads to  $\bar{\mathcal{G}} = \{g(\mathbf{z}) | g(\mathbf{z}) = a\mathbf{z}, a \neq 0\}$ . It is easy to verify that if  $g(\mathbf{z}) = a\mathbf{z}$ , then  $\bar{g}(\boldsymbol{\theta}) = a\boldsymbol{\theta}$ .

**Definition 2:** A group of transformations  $\mathcal{G}$  leaves a hypothesis-testing problem invariant if  $\mathcal{G}$  leaves both families of distributions  $\{f(\mathbf{z} | \boldsymbol{\theta}), \boldsymbol{\theta} \in \Theta_0\}$  and  $\{f(\mathbf{z} | \boldsymbol{\theta}), \boldsymbol{\theta} \in \Theta_1\}$  invariant.

It is easy to show that a hypothesis-testing problem remains invariant if and only if  $\bar{g}(\Theta_0) = \Theta_0$  and  $\bar{g}(\Theta_1) = \Theta_1$ . This condition expresses the requirement that the original dichotomy of  $\Theta$  is maintained for any transformation  $g$ . The detection problem of (49) is an invariant hypothesis test problem. To see this, note that for every  $\boldsymbol{\theta} = (\sigma_n^2, 0) \in \Theta_0$ ,  $\bar{g}(\boldsymbol{\theta}) = (a^2 \sigma_n^2, 0) \in \Theta_0$ . Further,  $\bar{g}(\Theta_0) \subseteq \Theta_0$ . In addition, for every  $\boldsymbol{\theta} = (\sigma_n^2, 0) \in \Theta_0$ ,  $\boldsymbol{\theta} = \bar{g}(\boldsymbol{\theta}/a^2)$ , hence  $\Theta_0 \subseteq \bar{g}(\Theta_0)$ . Consequently,  $\bar{g}(\Theta_1) = \Theta_1$ .

A decision rule is a mapping of the observation space  $\mathbf{z}$  to the decision space  $\{\mathcal{H}_0, \mathcal{H}_1\}$ . We say that a decision rule  $\phi: \mathbf{z} \rightarrow \{\mathcal{H}_0, \mathcal{H}_1\}$  is  $\mathcal{G}$  invariant if for all  $\mathbf{z}$  and  $g \in \mathcal{G}$

$$\phi(g(\mathbf{z})) = \phi(\mathbf{z}). \quad (50)$$

This condition reflects the requirement that the decision should be invariant to the group of transformations. For example, in our detection problem, we require that whether we observe  $\mathbf{z}$  or  $a\mathbf{z}$ , the decision should be the same. This requirement is manifested in mathematical terms by (50). The definition of the invariant decision rule leads to the following definition.

**Definition 3:** A function  $T(\mathbf{z})$  is said to be maximal invariant with respect to  $\mathcal{G}$  if (invariance)  $T(g(\mathbf{z})) = T(\mathbf{z})$ ,  $\forall \mathbf{z}$ , and  $\forall g \in \mathcal{G}$ , and (maximality)  $T(\mathbf{z}_1) = T(\mathbf{z}_2)$  implies  $\mathbf{z}_1 = g(\mathbf{z}_2)$  for some  $g \in \mathcal{G}$ .

The interpretation of maximal invariant statistic is beyond the scope of this paper, and the interested reader is referred to [33].

It is easy to verify that a maximal invariant statistic for our detection problem is given by

$$M(\mathbf{z}) = M(\|\mathbf{y}\|^2, \|\mathbf{x}\|^2) = \frac{\|\mathbf{x}\|^2}{\|\mathbf{y}\|^2}. \quad (51)$$

**Theorem 1:** A decision rule is  $\mathcal{G}$ -invariant if and only if it is a function of a maximal invariant statistic [27].

This theorem asserts that in searching for the optimal detector, we can limit our search only to functions of  $M(\mathbf{z})$  of (51).

In the following theorem, we develop the optimal scale invariant detector for our detection problem based on the theory described so far.

**Theorem 2:** Consider the detection problem of (49). An UMP scale invariant (UMPI) detector exists, and it is given by

$$T = \frac{\|\mathbf{x}\|^2}{\|\mathbf{y}\|^2} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{>}} \delta. \quad (52)$$

*Proof of Theorem 2:* The proof of the theorem is divided into two parts. In the first part, we will derive the optimal detector for two simple hypotheses, and in the second, we will demonstrate that the detector obtained in the first part is UMP.

Consider the following two simple hypotheses. Under the null hypothesis, the SNR is zero, while under the alternate hypothesis the SNR is a fixed value greater than zero. We are looking for the optimal invariant detector for deciding between these two hypothesis. According to Theorem 1 and the Neyman–Pearson Lemma, the optimal detector for this simple hypothesis testing problem is given by

$$T = \frac{f(M(\mathbf{z}) | \mathcal{H}_1)}{f(M(\mathbf{z}) | \mathcal{H}_0)} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{>}} \delta. \quad (53)$$

The maximal invariant test statistic,  $M(\mathbf{z})$ , is the result of dividing two independent chi-squared random variables. Consequently

$$\begin{aligned} M(\mathbf{z}) | \mathcal{H}_0 &\sim \frac{\chi_{2l(\mathbf{x})}^2}{\chi_{2L}^2} = \frac{l(\mathbf{x})}{L} \frac{\chi_{2l(\mathbf{x})}^2/2l(\mathbf{x})}{\chi_{2L}^2/2L} \\ &= \frac{l(\mathbf{x})}{L} \mathcal{F}_{2l(\mathbf{x}), 2L} \end{aligned} \quad (54)$$

$$\begin{aligned} M(\mathbf{z}) | \mathcal{H}_1 &\sim \frac{\sigma_n^2 + \mu}{\sigma_n^2} \frac{\chi_{2l(\mathbf{x})}^2}{\chi_{2L}^2} \\ &= \left(1 + \frac{\mu}{\sigma_n^2}\right) \frac{l(\mathbf{x})}{L} \mathcal{F}_{2l(\mathbf{x}), 2L} \end{aligned} \quad (55)$$

where and  $\mathcal{F}_{2l(\mathbf{x}), 2L}$  denotes an F-distributed random variable with  $2l(\mathbf{x})$  and  $2L$  degrees of freedom. Now

$$f(M(\mathbf{z}) | \mathcal{H}_0) = f_{\mathcal{F}_{2l(\mathbf{x}), 2L}}\left(\frac{M(\mathbf{z})L}{l(\mathbf{x})}\right) \quad (56)$$

$$f(M(\mathbf{z}) | \mathcal{H}_1) = f_{\mathcal{F}_{2l(\mathbf{x}), 2L}}\left(\frac{M(\mathbf{z})L}{l(\mathbf{x})(1 + \text{SNR})}\right). \quad (57)$$

where  $f_{\mathcal{F}_{2l(\mathbf{x}), 2L}}(\cdot)$  is the pdf of an  $\mathcal{F}_{2l(\mathbf{x}), 2L}$  random variable.

Since the pdf of the  $F$  distribution is monotonically decreasing,  $f(M(\mathbf{z}) | \mathcal{H}_1, \text{SNR})/f(M(\mathbf{z}) | \mathcal{H}_0)$  is monotonically increasing. Consequently, (53) is equivalent to the following detector:

$$T = M(\mathbf{z}) \underset{\mathcal{H}_1}{\overset{\mathcal{H}_0}{>}} \delta. \quad (58)$$

Since both the optimal test statistic  $T$  and  $\delta$  are independent of the unknown parameters, (58) is a uniformly most powerful scale invariant detector [27].  $\square$

### C. Discussion

Theorem 2 describes the optimal detector for the general detection problem in (49). In this subsection, we discuss this general problem in the context of the phased-array, MIMO, and

MISO radar systems. We compare the various systems based on their optimal scale invariant detector's SNR.

Denote by  $\rho = E/\sigma_n^2$  the average received SNR. By using the same method leading to (42), (43), and (44), respectively, we have

$$\beta_{\text{MIMO-UMPI}} = \beta_{\text{MIMO}} \frac{(2L-4)}{(2MN+2L-2)} \quad (59)$$

$$\beta_{\text{phased array-UMPI}} = \beta_{\text{phased array}} \frac{2L-4}{2L} \quad (60)$$

$$\beta_{\text{MISO-UMPI}} = \beta_{\text{MISO}} \frac{2L-4}{2L+2M-2}. \quad (61)$$

From the expressions, it is evident that the number  $L$  of noise samples has great influence on the UMPI detectors' SNR. In each case, the SNR is equal to the optimal detector's SNR multiplied by a system dependent factor smaller than one. This factor represents the degradation in performance due to the unknown parameters, and it approaches one as the number of noise samples approaches infinity. This should not come as a surprise because, when the number of noise samples is high, the noise level can be estimated with very small error.

The number of independent noise samples we can obtain depends on the number of transmitters and receivers, the system complexity, and the model used. In clutter limited systems the only way to obtain samples of the noise process is to sample the output of the filters which are matched to the transmitted signals at locations (delays) that do not contain targets. In a typical radar system, between 8 and 16 such locations are used for estimating the noise level. Consider one such location [17]. In our proposed MIMO radar system, we can sample each of the  $MN$  matched filters at that delay and obtain  $MN$  independent samples of the noise process. Similar reasoning applies for the MISO system as well. In the phased-array system, at each such location, only  $N$  independent samples can be obtained because only one signal is transmitted by the transmitting array. Therefore, in practical radar systems we can expect about  $10MN$  independent samples of the noise process if MIMO or MISO systems are used, and about  $10N$  samples if a phased-array radar systems are used.

Fig. 3 depicts the performance of the phased-array, MIMO, and MISO radar systems' UMPI detectors for  $M = N = 4$ , and that the noise level is constant across the array. In addition, the number of independent samples of the noise process was set to either  $8MN$  or  $MN$ . Fig. 3 demonstrates the validity of our theoretical results. We can see that when the number of noise samples is large, the degradation in SNR is negligible. However, if the number of noise samples is very small, the degradation in the SNR of the MIMO and MISO systems is larger than that of the phased-array system.

## V. NUMERICAL EXAMPLE

In this section, we consider several numerical examples that compare the performance of the various systems. These examples also validate the analysis conducted in the previous sections. In our first example we consider systems having four receiving antennas and two transmitting antennas, that is  $N = 4, M = 2$ . In this example, we consider the MIMO, MISO, and phased-array systems. Fig. 4 depicts the probability of miss of the optimal detectors for both known and unknown noise levels

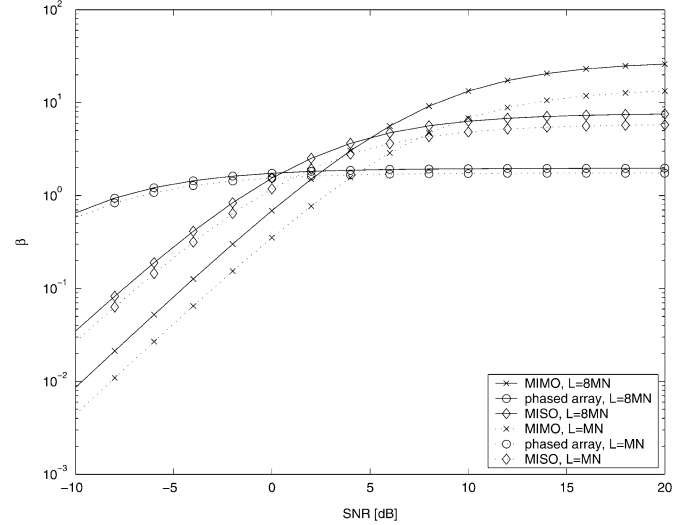


Fig. 3. Optimal detector's SNR for various systems.  $M = N = 4$ .

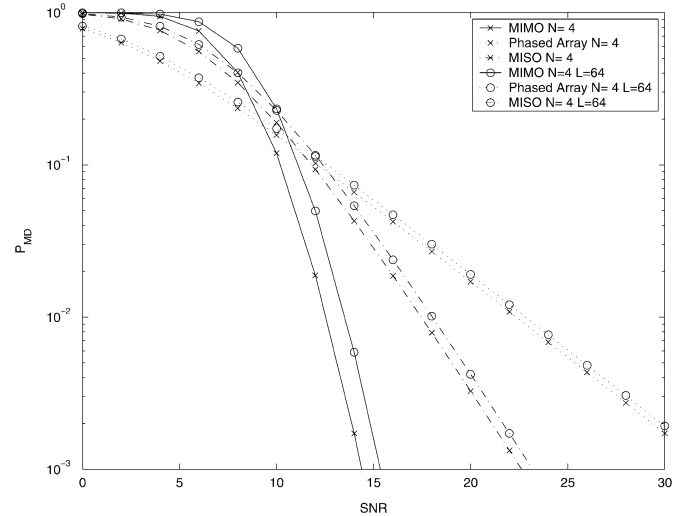


Fig. 4. Probability of miss-detection as a function of the SNR,  $N = 4, M = 2$ .

as a function of the SNR. The probability of false alarm was fixed at  $P_{FA} = 10^{-6}$ . We assume that when the noise level is unknown the receiver obtains 64 independent samples of the noise process.

At low SNR, the phased-array system outperforms the MIMO system, while at high SNR the MIMO system outperforms the phased-array system. However, we can see that the MIMO radar system outperforms the phased-array system when the probability of miss is about 20%. We can also see from the results that the MISO system is, in a sense, a compromise between the MIMO and phased-array systems. Therefore, at low SNR, the MISO system outperforms the MIMO system, while at high SNR, the MISO system outperforms the phased-array system.

Fig. 5 depicts the receiver operating curves of the MIMO, MISO, and phased-array radar systems under known noise level and unknown noise level conditions. The average received SNR was set at  $\rho = 10$  dB. This figure establishes the advantage of the MIMO radar system over the phased-array radar system. For known noise level, the MIMO system outperforms the phased-

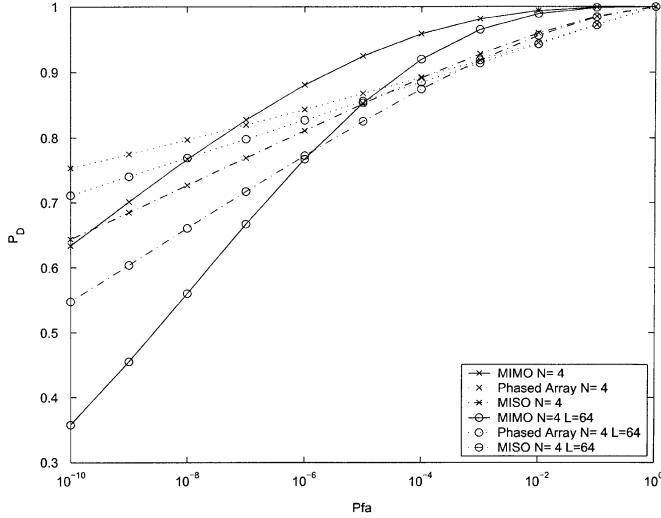


Fig. 5. Receiver's operating curves,  $N = 4$ ,  $M = 2$ .

array system at probability of false alarm greater than  $10^{-7.5}$ , whereas, for an unknown noise level, the MIMO system outperforms the phased-array system at a probability of false alarm greater than  $10^{-5}$ . Again, the MISO system offers a compromise between the two systems.

In order to fully appreciate the performance advantage of the MIMO radar system, one has to understand the limitation of the approach we used to compare the various systems. MIMO radar systems illuminate the space uniformly. However, phased-array radar system cohere a beam that needs to scan the whole space. Phased array that is equipped with  $M$  transmitting antennas can create a beam with beamwidth  $2\pi/M$ , and hence it can realize a processing gain of  $M$  (in linear scale). However, since the transmitter needs to scan the whole space in, say,  $T$  seconds, it can illuminate the target for only  $T/M$  seconds. Therefore, during the time it takes the phased-array system to scan the whole space, the MIMO and MISO system can continue to illuminate the target and realize processing gain of  $M$  (in linear scale) above and beyond the results reported. The reader who is interested in comparing the performance of the various systems based on the same surveillance volume should either penalize the phased-array system by  $10 \log_{10} M$  or compensate for the time it takes the system to scan the whole space. It is fair to point out that while it offers many advantages, the setup of a multistatic MIMO radar is considerably more complex and costly than monostatic radar.

## VI. SUMMARY AND CONCLUDING REMARKS

In this paper, we introduced the concept of MIMO radar. This concept differs substantially from current regimes in which closely spaced antenna arrays are used. With closely spaced antenna elements, it is possible to cohere a beam toward a direction in space and to realize a coherent processing gain. However, these systems are prone to severe target fading, and hence they may suffer considerable performance degradation. In contrast, MIMO radar can not cohere a beam toward a certain direction in space, and hence it can not realize any processing gain. However, MIMO radar exploits that target angular spread to combat

target fading. MIMO radar is composed of many independent radars, each of which sees a different aspect of the target, enabling the MIMO radar to exploit spatial diversity to overcome target fading.

In addition, we extended the current models used in radar theory to account for both range and angular spreads, and sparse antenna arrays. While, for closely spaced arrays, this model is equivalent to the current existing models, for sparse antenna arrays, this model provides for the first time a simple method for modeling the signals received from complex targets. In addition, this model offers insights into the advantages and disadvantages of each system considered in this paper. From the model, we can learn how MIMO radar can exchange coherent processing gain for diversity gain.

More important, we investigated and compared the inherent performance limitations of both conventional phased-array radars and the newly proposed radars. We derived the optimal detectors for both the conventional and the newly proposed systems when the parameters are either known or unknown. We demonstrated that the MIMO radar outperforms the conventional phased-array radar whenever the probability of detection is at a reasonable level, say above 80%.

It should be noted that spatial diversity is not the only type of diversity that one exploit. Temporal as well as frequency diversity can be used as well. In a subsequent publication, we will investigate interactions between these diversity mechanisms and spatial diversity.

It is the current state of understanding that one should always try to increase the SNR by coherent combining. In this paper, we suggest to deviate from this regime and to balance between coherent processing and diversity. This paper provides a partial answer to the problem we refer to as “detection with unit energy.” This problem is defined as follows: Given a fixed amount of energy and a scintillating target, should we cohere a beam toward the target and see only one aspect of the target or observe several aspects of the target with reduced power? Another aspect of the same problem is the need to decide whether to transmit one strong pulse or to divide our energy between several pulses.

The performance improvement achieved by MIMO radar is not limited to detection. In subsequent papers, we will provide a detailed analysis which demonstrates the superiority of MIMO radar in many aspects over the conventional phased-array radar. We show that MIMO radar enjoys lower range, location, angle of arrival, and Doppler estimation errors. A preliminary analysis that demonstrates these results can be found in [25] and [35].

## APPENDIX I

### PROOF OF LEMMA 1

In this section we compute the likelihood ratio test (LRT) detector for multiple-input multiple-output (MIMO) radar systems. We start by deriving the pdf of the received measurements under both the null and alternative hypotheses. Consider first the alternative hypothesis found in (62), shown at the top of the next page, where  $\mathbf{x}$  is an  $NM \times 1$  vector such that  $[\mathbf{x}]_{iN+j} \triangleq \int r_i(t) s_j(t - \tau) dt$ ,  $\alpha$  is the  $NM \times 1$  vector that contains the fading coefficients, and in (a) we use the fact that  $\int s_j(t) s_i(t) dt = \delta_{ij}$ .

$$\begin{aligned}
f(\mathbf{r}(t) | \mathcal{H}_1) &= \int f(\mathbf{r}(t) | \mathcal{H}_1, \boldsymbol{\alpha}) f(\boldsymbol{\alpha}) d\boldsymbol{\alpha} = \int c e^{-\frac{1}{\sigma_n^2} \int \|\mathbf{r}(t) - \mathbf{H} \sqrt{\frac{E}{M}} \mathbf{s}(t-\tau)\|^2 dt} f(\boldsymbol{\alpha}) d\boldsymbol{\alpha} \\
&= \int c e^{-\frac{1}{\sigma_n^2} \sum_i \int |r_i(t) - \sqrt{\frac{E}{M}} \sum_{j=1}^M \alpha_{ij} s_j(t-\tau)|^2 dt} f(\boldsymbol{\alpha}) d\boldsymbol{\alpha} \\
&\stackrel{(a)}{=} \int c e^{-\frac{1}{\sigma_n^2} \sum_i \int |r_i(t)|^2 dt - \sqrt{\frac{E}{M}} \sum_{j=1}^M \alpha_{ij}^* \int r(t) s_j^*(t-\tau) dt - \sqrt{\frac{E}{M}} \sum_{j=1}^M \alpha_{ij} \int r^*(t) s_j(t-\tau) dt + \frac{E}{M} \sum_{j=1}^M |\alpha_{ij}|^2} f(\boldsymbol{\alpha}) d\boldsymbol{\alpha} \\
&= c' e^{-\frac{\int \|\mathbf{r}(t)\|^2 dt}{\sigma_n^2}} \int e^{-\frac{1}{\sigma_n^2} (-\sqrt{\frac{E}{M}} \boldsymbol{\alpha}^H \mathbf{x} - \sqrt{\frac{E}{M}} \mathbf{x}^H \boldsymbol{\alpha} + \|\sqrt{\frac{E}{M}} \boldsymbol{\alpha}\|^2)} e^{-\|\boldsymbol{\alpha}\|^2} d\boldsymbol{\alpha} \\
&= c' e^{-\frac{\int \|\mathbf{r}(t)\|^2 dt}{\sigma_n^2} + \frac{\frac{E}{M} \|\mathbf{x}\|^2}{\sigma_n^2(\sigma_n^2 + \frac{E}{M})}} \int e^{-\frac{1}{\sigma_n^2} \|\sqrt{\sigma_n^2 + \frac{E}{M}} \boldsymbol{\alpha} - \frac{\sqrt{\frac{E}{M}} \mathbf{x}}{\sqrt{\sigma_n^2 + \frac{E}{M}}}\|^2} d\boldsymbol{\alpha} = c'' e^{-\frac{\int \|\mathbf{r}(t)\|^2 dt}{\sigma_n^2} + \frac{\frac{E}{M} \|\mathbf{x}\|^2}{\sigma_n^2(\sigma_n^2 + \frac{E}{M})}}. \quad (62)
\end{aligned}$$

$$\begin{aligned}
f(\mathbf{r}(t) | \mathcal{H}_1) &= \int f(\mathbf{r}(t) | \mathcal{H}_1, \boldsymbol{\alpha}) f(\boldsymbol{\alpha}) d\boldsymbol{\alpha} = \int c e^{-\frac{1}{\sigma_n^2} \int \|\mathbf{r}(t) - \sqrt{\frac{E}{M}} \boldsymbol{\alpha} \mathbf{a}(\theta) \mathbf{b}^H(\theta') \tilde{\mathbf{b}} \mathbf{s}(t-\tau)\|^2 dt} f(\boldsymbol{\alpha}) d\boldsymbol{\alpha} \\
&= c e^{-\frac{\int \|\mathbf{r}(t)\|^2 dt}{\sigma_n^2}} \int e^{-\frac{1}{\sigma_n^2} (-2\sqrt{\frac{E}{M}} \Re(\boldsymbol{\alpha} \mathbf{b}^H(\theta') \tilde{\mathbf{b}} \int \mathbf{r}^H(t) \mathbf{a}(\theta) \mathbf{s}(t-\tau) dt) + \frac{E}{M} |\boldsymbol{\alpha}|^2 \|\tilde{\mathbf{b}}^H \mathbf{b}(\theta) \mathbf{a}(\theta)\|^2 + |\boldsymbol{\alpha}|^2)} d\boldsymbol{\alpha} \\
&= c' e^{-\frac{\int \|\mathbf{r}(t)\|^2 dt}{\sigma_n^2} - \frac{E \|\mathbf{b}(\theta') \tilde{\mathbf{b}}\|^2 \|\int \mathbf{r}^H(t) \mathbf{a}(\theta) \mathbf{s}(t-\tau) dt\|^2}{M \sigma_n^2 (1 + \frac{E}{M} \|\mathbf{b}(\theta') \tilde{\mathbf{b}}\|^2)}} = c' e^{-\frac{\int \|\mathbf{r}(t)\|^2 dt}{\sigma_n^2} - \frac{E \|\mathbf{b}(\theta') \tilde{\mathbf{b}}\|^2 \|\mathbf{x}\|^2}{M \sigma_n^2 (1 + \frac{E}{M} \|\mathbf{b}(\theta') \tilde{\mathbf{b}}\|^2)}}. \quad (65)
\end{aligned}$$

The probability density function (pdf) of the received measurements under the null hypothesis is given by

$$f(\mathbf{r}(t) | \mathbf{H}_0) = c e^{-\frac{\int \|\mathbf{r}(t)\|^2 dt}{\sigma_n^2}}. \quad (63)$$

Combining (23), (62), and (63) results in the following detector:

$$T = \log f(\mathbf{r}(t) | \mathbf{H}_1) - \log f(\mathbf{r}(t) | \mathcal{H}_0) = \|\mathbf{x}\|^2 \stackrel{\mathbf{H}_1}{<}_{\mathbf{H}_0} \delta. \quad (64)$$

## APPENDIX II PROOF OF LEMMA 2

In order to prove Lemma 2, we need to derive the probability density function (pdf) of the received measurements under both the null and alternative hypotheses. We first consider the alternative hypothesis, found in (65), shown at the top of the page, where  $\mathbf{x} = \int \mathbf{r}^H(t) \mathbf{a}(\theta) \mathbf{s}(t-\tau) dt$ .

The pdf of the received measurements under the null hypothesis is given by (63). Combining (23), (63), and (65), results in the following detector:

$$\begin{aligned}
T &= \log f(\mathbf{r}(t) | \mathbf{H}_1) - \log f(\mathbf{r}(t) | \mathcal{H}_0) \\
&= \int \mathbf{r}^H(t) \mathbf{a}(\theta) \mathbf{s}(t-\tau) dt \stackrel{\mathbf{H}_1}{>}_{\mathbf{H}_0} \delta. \quad (66)
\end{aligned}$$

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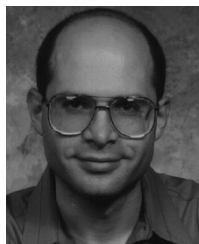
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